

Practical_12

a.

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udayasri@SURABHIUDAYASRI:~/OSLab$ gedit practical12a.c
udayasri@SURABHIUDAYASRI:~/OSLab$ gcc practical12a.c -o practical12a
udayasri@SURABHIUDAYASRI:~/OSLab$ ./practical12a
PRODUCER-CONSUMER USING SEMAPHORES
=====

Buffer Size : 2
Items       : 10000
Time        : 0.1754 seconds
Throughput  : 57005.07 items/second
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Buffer Size : 5
Items       : 10000
Time        : 0.0716 seconds
Throughput  : 139672.61 items/second
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Buffer Size : 10
Items       : 10000
Time        : 0.0309 seconds
Throughput  : 323206.21 items/second
-----

Buffer Size : 20
Items       : 10000
Time        : 0.0157 seconds
Throughput  : 638732.75 items/second
-----
udayasri@SURABHIUDAYASRI:~/OSLab$ |
```

b.

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udayasri@SURABHIUDAYASRI:~/OSLab$ gedit practical12b.c
udayasri@SURABHIUDAYASRI:~/OSLab$ gcc practical12b.c -o practical12b
udayasri@SURABHIUDAYASRI:~/OSLab$ ./practical12b
DEADLOCK DEMONSTRATION
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Thread 1: Trying to lock Resource A
Thread 1: Locked Resource A
Thread 2: Trying to lock Resource B
Thread 2: Locked Resource B
Thread 1: Trying to lock Resource B
Thread 2: Trying to lock Resource A
```

c.

```
udayasri@SURABHIUDAYASRI:~/OSLab$ gedit practical12c.c
udayasri@SURABHIUDAYASRI:~/OSLab$ gcc practical12c.c -o practical12c
udayasri@SURABHIUDAYASRI:~/OSLab$ ./practical12c
DEADLOCK PREVENTION USING RESOURCE ORDERING
=====

Thread 1: Locking Resource A
Thread 1: Locked Resource A
Thread 2: Locking Resource A
Thread 1: Locking Resource B
Thread 1: Locked Resource B
Thread 1: Using both resources
Thread 1: Released resources
Thread 2: Locked Resource A
Thread 2: Locking Resource B
Thread 2: Locked Resource B
Thread 2: Using both resources
Thread 2: Released resources

No deadlock occurred.
udayasri@SURABHIUDAYASRI:~/OSLab$ |
```